Algebra 1: Polynomials

Homework$_3$*Level*

Name: ____________________

Date: ____________________

Question 1: Simplify the expression $x^2 + 3x + 2$

A) $x^2 + 5x + 2$
B) $x^2 + 3x + 1$
C) $x^2 + 3x$
D) $x + 2$
E) $x^2 + 3x + 2$

Question 2: Factor the expression $x^2 - 4$

A) $x^2 - 2x + 2x - 4$
B) $x^2 + 2x - 2x - 4$
C) $(x - 2)(x + 2)$
D) $(x + 2)(x - 2)$
E) $x^2 - 2x + 2$

Question 3: Solve for x: $2x + 5 = 11$

A) $x = 2$
B) $x = 3$
C) $x = 4$
D) $x = 5$
E) $x = 6$

Question 4: Find the value of x for which $x^2 + 2x - 6 = 0$

A) $x = -1 pmsqrt7$
B) $x = 1 pmsqrt7$
C) $x = -3, x = 2$
D) $x = 3, x = -2$
E) $x = 2, x = -3$

Question 5: Given the equation $x^2 - 5x + 6 = 0$, find the values of x

A) $x = 2, x = 3$
B) $x = -2, x = -3$
C) $x = 2, x = -3$
D) $x = -2, x = 3$
E) $x = 1, x = 6$

Question 6: Solve the equation $x^2 + 4x + 4 = 0$

A) $x = -2$
B) $x = 2$
C) $x = -1$
D) $x = 1$
E) $x = 0$

Question 7: The expression $x^3 - 6x^2 + 11x - 6$ can be factored as

A) $(x-1)(x-2)(x-3)$
B) $(x+1)(x+2)(x+3)$
C) $(x-1)(x+2)(x+3)$
D) $(x+1)(x-2)(x-3)$
E) $(x-2)(x-3)(x+1)$

Question 8: Given $f(x) = x^3 - 2x^2 - 5x + 6$, find $f(-1)$

A) -4
B) -2
C) 0
D) 2
E) 4

Question 9: The polynomial $x^3 + 2x^2 - 7x - 12$ can be divided by $x + 3$ with a remainder of

A) 0
B) 1
C) 2
D) 3
E) -3

Question 10: Prove that $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$ by showing all the steps. Use LaTeX to format your expressions, e.g. $x^2 + 2x + 1$.

Chef's Tips (Hints):

- To factor expressions, look for common factors first and then use methods like grouping or the quadratic formula.
- When solving equations, try to isolate the variable by performing the inverse operations.
- When dividing polynomials, use long division or synthetic division and be sure to include the remainder.